

FLAG ISLAND, MN, USA

WMO: 727467

Lat: 49.318N

Lon: 94.903W

Elev: 329

StdP: 97.43

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04926

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.9	-27.7	-33.9	0.2	-31.0	-32.0	0.2	-28.0	12.9	-11.4	12.0	-12.2	2.4	310	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	27.4	20.3	26.2	19.7	24.7	19.2	22.4	25.7	21.3	24.7	20.2	23.5	3.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.3	16.6	25.1	20.0	15.3	24.0	18.9	14.2	23.0	67.9	25.8	63.4	24.7	59.7	23.7	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	10.9	9.4	-34.4	29.8	2.6	1.4	-36.3	30.8	-37.8	31.6	-39.2	32.4	-41.1	33.5
			-34.4	23.5	2.4	1.0	-36.2	24.3	-37.6	24.9	-39.0	25.5	-40.7	26.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-14.5	-13.5	-5.4	2.9	10.7	17.1	20.5	19.3	14.5	6.7	-1.8	-11.1	(6)
	DBStd	13.53	7.85	7.01	7.05	5.26	4.69	3.51	2.91	3.04	4.27	4.64	5.73	7.25	(7)
	HDD10.0	3306	758	658	477	218	49	1	0	0	10	125	354	656	(8)
	HDD18.3	5456	1017	892	735	464	239	64	13	26	127	362	603	914	(9)
	CDD10.0	1067	0	0	0	4	70	213	325	288	146	22	0	0	(10)
	CDD18.3	175	0	0	0	0	2	26	80	54	13	0	0	0	(11)
	CDH23.3	612	0	0	0	0	7	73	322	189	21	0	0	0	(12)
	CDH26.7	68	0	0	0	0	1	6	38	23	1	0	0	0	(13)
	WSAvg	4.2	4.2	4.0	4.0	4.1	4.4	3.8	3.8	3.5	4.5	5.2	5.2	4.3	(14)
	PrecAvg	695	30	24	32	38	82	100	105	75	81	59	40	31	(15)
(16) Precipitation	PrecMax	865	55	48	59	72	158	205	200	159	152	128	101	56	(16)
	PrecMin	461	11	6	11	16	31	48	38	4	26	11	12	12	(17)
	PrecStd	115	11	10	14	16	37	42	40	38	38	27	23	11	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	4.1	12.3	17.5	24.0	27.5	29.1	28.9	26.0	19.2	12.0	2.9	(19)
		MCWB	1.1	1.6	7.4	9.2	15.7	18.6	21.0	21.2	19.3	14.2	8.1	1.6	(20)
	2%	DB	1.2	2.3	7.4	14.0	22.1	25.9	27.6	27.1	23.2	17.4	8.8	1.3	(21)
		MCWB	0.3	0.9	4.2	7.7	14.6	17.8	20.5	20.8	18.1	13.0	6.6	0.3	(22)
	5%	DB	-1.2	0.1	5.8	12.3	19.1	23.7	26.4	25.9	22.3	14.9	7.3	0.0	(23)
		MCWB	-1.5	-0.8	3.1	7.0	13.9	17.8	19.8	20.2	17.8	11.2	5.2	-0.5	(24)
	10%	DB	-3.0	-2.5	3.0	10.2	17.5	22.4	25.1	23.9	21.0	12.8	6.0	-1.4	(25)
		MCWB	-3.9	-3.5	1.3	6.0	12.8	17.2	19.5	19.1	17.4	9.9	4.0	-1.6	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	2.4	7.9	10.3	18.6	21.7	23.8	23.7	20.9	15.4	8.6	1.9	(27)
		MCDB	3.2	3.7	10.3	15.1	21.6	25.0	27.1	26.7	24.1	19.1	10.6	2.8	(28)
	2%	WB	0.2	0.8	5.0	9.1	16.4	20.0	22.7	22.3	19.6	13.7	7.0	0.4	(29)
		MCDB	1.0	1.9	7.5	13.2	19.6	23.2	26.0	25.6	22.4	16.6	8.3	1.0	(30)
	5%	WB	-1.8	-1.0	3.0	7.6	14.8	19.0	21.7	21.1	18.7	12.0	5.4	-0.7	(31)
		MCDB	-1.5	-0.1	4.9	11.3	18.3	22.1	25.0	24.6	21.4	14.6	7.3	-0.2	(32)
	10%	WB	-4.1	-3.4	1.7	6.0	13.2	17.9	20.5	19.9	17.5	10.2	4.0	-2.1	(33)
		MCDB	-3.1	-2.4	3.3	9.8	16.8	20.9	24.0	23.2	19.8	12.7	5.8	-1.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	10.5	10.0	8.4	7.4	6.6	6.8	6.6	6.3	5.3	5.1	7.0	(35)
		MCDBR	9.3	9.9	10.3	11.2	10.0	8.1	7.8	7.9	7.7	7.5	6.2	7.5	(36)
		MCWBR	8.4	8.4	7.4	6.4	5.1	4.5	4.6	4.7	4.7	4.6	4.4	6.7	(37)
	5% WB	MCDBR	9.1	9.3	9.4	10.3	8.2	7.2	6.8	7.0	6.7	6.9	5.9	7.0	(38)
		MCWBR	8.4	8.0	7.0	6.5	5.0	4.8	4.8	4.7	4.6	4.8	4.3	6.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.250	0.268	0.294	0.335	0.352	0.373	0.378	0.370	0.346	0.304	0.272	0.248	(40)
	taud		2.323	2.286	2.389	2.422	2.441	2.405	2.416	2.453	2.502	2.585	2.487	2.353	(41)
	Ebn at Noon		821	888	924	914	908	886	876	868	857	848	797	773	(42)
	Edh at Noon		67	89	97	105	108	113	110	101	87	68	58	57	(43)
(44) All-Sky Solar Radiation	RadAvg		1.32	2.49	3.69	4.76	5.38	5.63	5.98	4.96	3.66	2.12	1.16	0.93	(44)
	RadStd		0.14	0.20	0.36	0.46	0.37	0.45	0.32	0.37	0.31	0.30	0.15	0.10	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page